

Tutorial: Phase 3 - Frontend Development

Introduction

This tutorial provides step-by-step instructions for developing the **frontend UI** for the **Pandemic Resilience System (PRS)**. The frontend will include user authentication, a data visualization dashboard, and API integration for real-time insights.

Objectives:

Develop HTML/CSS-based UI for:

- Dashboard (Vaccination insights, supply chain)
- User login & registration
- Data visualization (Charts & graphs)
 - Implement JavaScript (AJAX) to fetch API data
 - Integrate **Chart.js** for data visualization:
- Bar chart (Vaccination distribution)
- Pie chart (Vaccine types per region)
- Line chart (Trend of vaccinations over time)
- Make UI mobile-responsive

Deliverables:

- Functional frontend with API integration
- Interactive charts displaying real-time data
- Responsive design with user authentication

Step 1: Set Up Frontend Development Environment

Install Required Tools

- **Text Editor/IDE:** VS Code, Sublime Text, or Atom
- **Web Server:** XAMPP (for local PHP support)
- **Libraries:**
 - [Chart.js](#) for data visualization
 - Bootstrap (for UI responsiveness)

Folder Structure

```
/prs-frontend/  
├── index.html    # Dashboard Page  
├── login.html    # User Login Page  
├── register.html # User Registration Page  
├── css/  
│   └── styles.css # Stylesheet  
├── js/  
│   └── scripts.js # JavaScript for API calls & Chart.js  
├── assets/      # Images & icons  
└── api/         # API integration files
```

Step 2: Build User Authentication Pages

Create Login Page (login.html)

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <title>Login | PRS</title>  
  <link rel="stylesheet" href="css/styles.css">  
</head>  
<body>  
  <div class="login-container">  
    <h2>Login</h2>  
    <form id="loginForm">  
      <input type="email" id="email" placeholder="Email" required>  
      <input type="password" id="password" placeholder="Password"  
required>  
      <button type="submit">Login</button>  
    </form>  
    <p>New user? <a href="register.html">Register here</a></p>  
  </div>  
  <script src="js/scripts.js"></script>  
</body>  
</html>
```

Create User Registration Page (register.html)

```
<form id="registerForm">  
  <input type="text" id="full_name" placeholder="Full Name" required>  
  <input type="email" id="email" placeholder="Email" required>  
  <input type="password" id="password" placeholder="Password" required>  
  <button type="submit">Register</button>  
</form>
```

Step 3: Create the Dashboard (index.html)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Dashboard | PRS</title>
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
</head>
<body>
  <h2>Vaccination Insights</h2>
  <canvas id="barChart"></canvas>
  <canvas id="pieChart"></canvas>
  <canvas id="lineChart"></canvas>
  <script src="js/scripts.js"></script>
</body>
</html>
```

Step 4: Fetch API Data & Display Charts

JavaScript (scripts.js)

```
document.addEventListener("DOMContentLoaded", function() {
  fetch('api.php?action=get_vaccination_data')
    .then(response => response.json())
    .then(data => {
      const labels = data.map(item => item.full_name);
      const doses = data.map(item => item.dose_number);
      const vaccineCounts = {};
      data.forEach(item => {
        vaccineCounts[item.vaccine_name] =
(vaccineCounts[item.vaccine_name] || 0) + 1;
      });

      const pieLabels = Object.keys(vaccineCounts);
      const pieData = Object.values(vaccineCounts);

      // Bar Chart
      const ctx = document.getElementById('barChart').getContext('2d');
      new Chart(ctx, {
        type: 'bar',
        data: {
          labels: labels,
          datasets: [{
            label: 'Number of Doses',
            data: doses,

```

```

        backgroundColor: 'rgba(75, 192, 192, 0.2)',
        borderColor: 'rgba(75, 192, 192, 1)',
        borderWidth: 1
      }],
    },
    options: { responsive: true }
  });

  // Pie Chart
  const pieCtx =
document.getElementById('pieChart').getContext('2d');
  new Chart(pieCtx, {
    type: 'pie',
    data: {
      labels: pieLabels,
      datasets: [{
        data: pieData,
        backgroundColor: ['#ff6384', '#36a2eb', '#ffce56']
      }]
    }
  });
})
.catch(error => console.error('Error fetching data:', error));
});

```

Step 5: Make UI Mobile-Responsive (styles.css)

```

body { font-family: Arial, sans-serif; }
canvas { max-width: 100%; }
@media (max-width: 768px) {
  canvas { width: 100%; }
}

```

Deliverables

- Fully functional frontend with API integration
- Interactive charts displaying real-time data
- Responsive design with user authentication